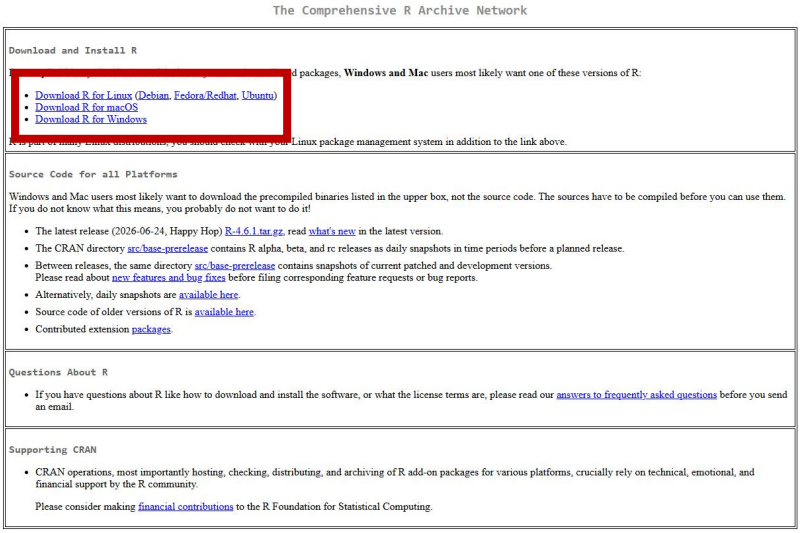
User Guide: Toolkit for Publication-Ready Figures

This is an open-source, locally run R Shiny workflow designed to prepare biomedical figures according to official PubMed Central (PMC) technical standards. Running directly on your personal computer via R and RStudio, it requires no paid software and ensures patient/clinical image security by keeping all data local.

1. System Setup & Launching the Suite

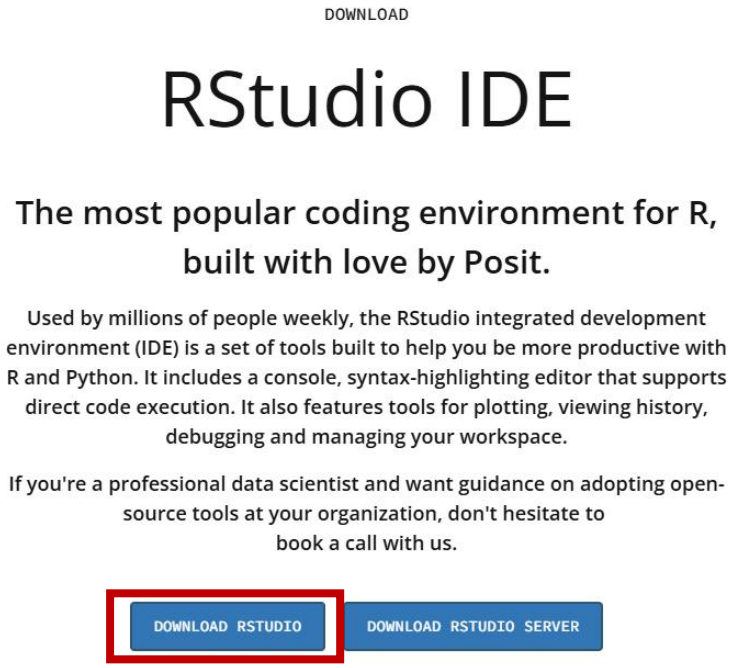
> Install R

Download and install R from the CRAN repository. https://cran.r-project.org



> Install RStudio

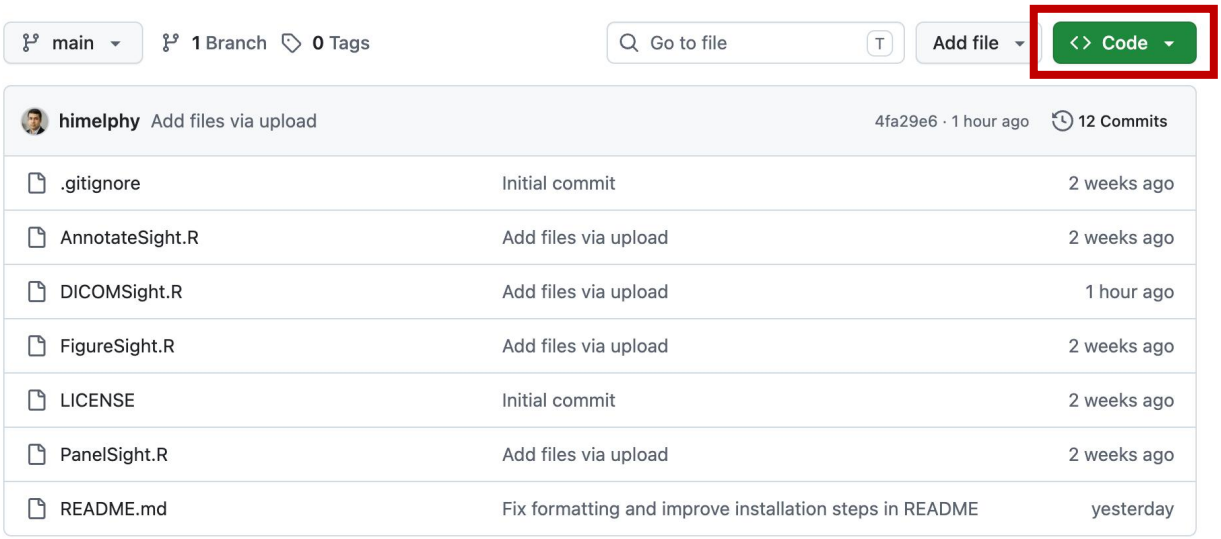
Download and install the desktop version from Posit. https://posit.co/downloads



License: CC-BY 4.0

> Get the Applications

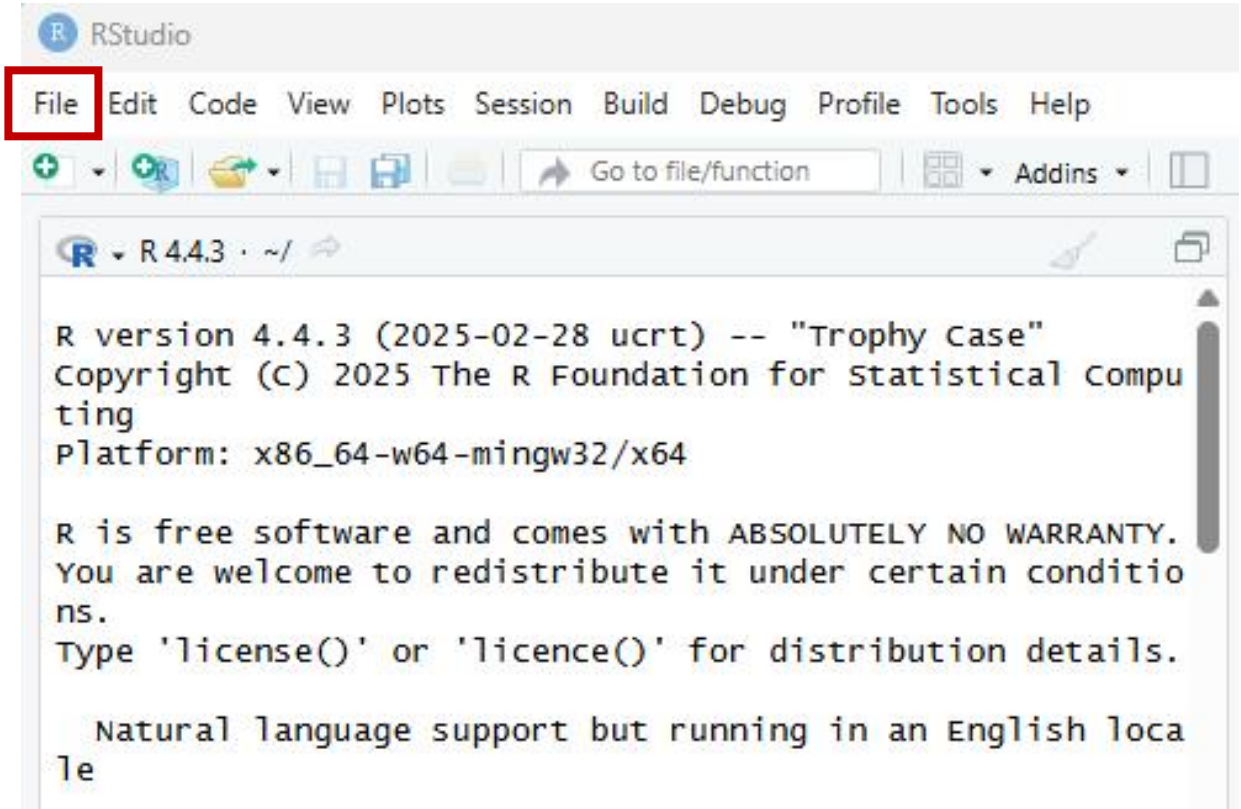
Visit GitHub: FigureToolkit, click Code, and select Download ZIP. Extract the .zip archive on your computer. https://github.com/himelphy/FigureToolkit



2. Launching an Application

> Open RStudio

Double-click on the RStudio icon



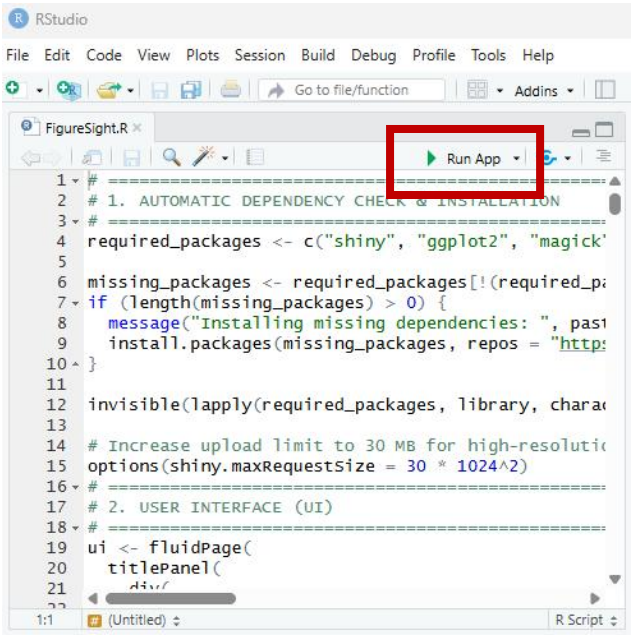
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> Open the specific script you need

To open any of the applications (FigureSight.R, AnnotateSight.R, or PanelSight.R), go to File > Open File

> Run Application

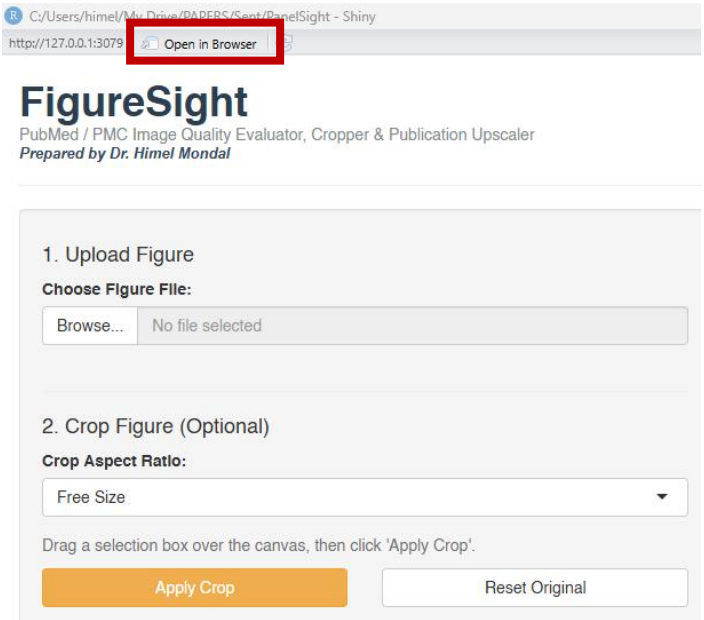
Click the Run App button in the top-right corner of the editor window.



Dependency Check: On the first run, each application script automatically checks for and installs all missing R package dependencies. Allow all.

> Interface

The application interface will open in a new RStudio window or your default web browser.



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3. Application Modules

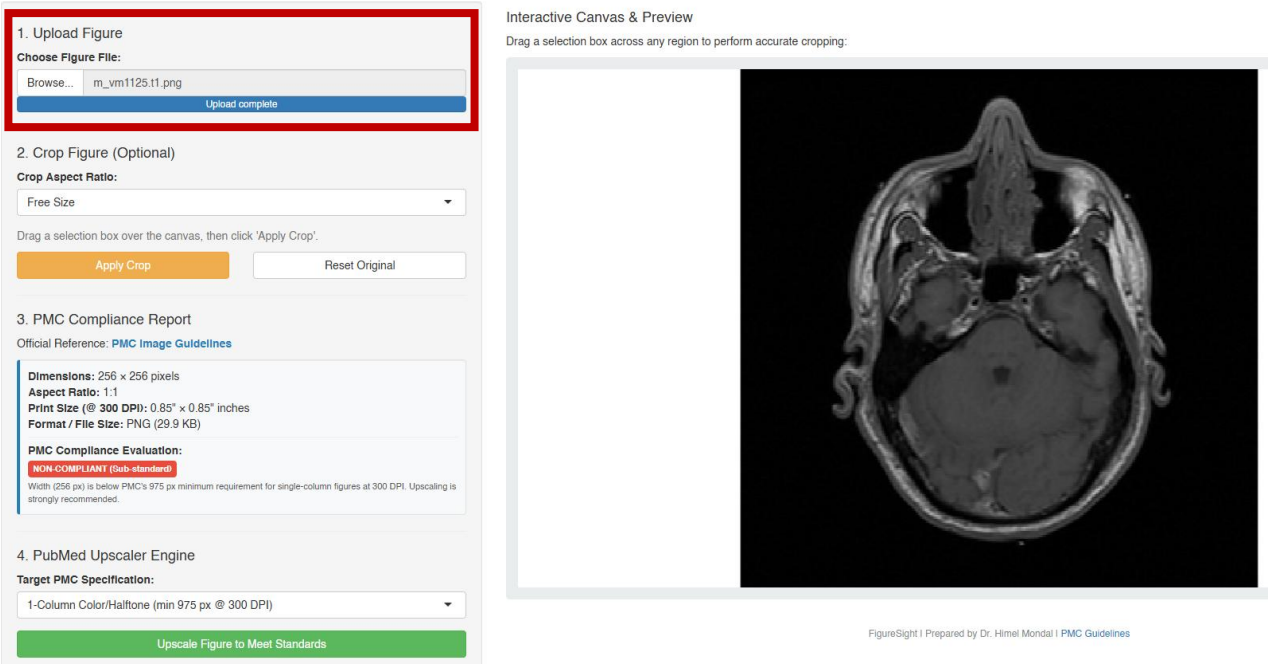
> Module 1: FigureSight

Audit raw figures against PMC guidelines, crop regions, and adjust pixel dimensions.

Functions:

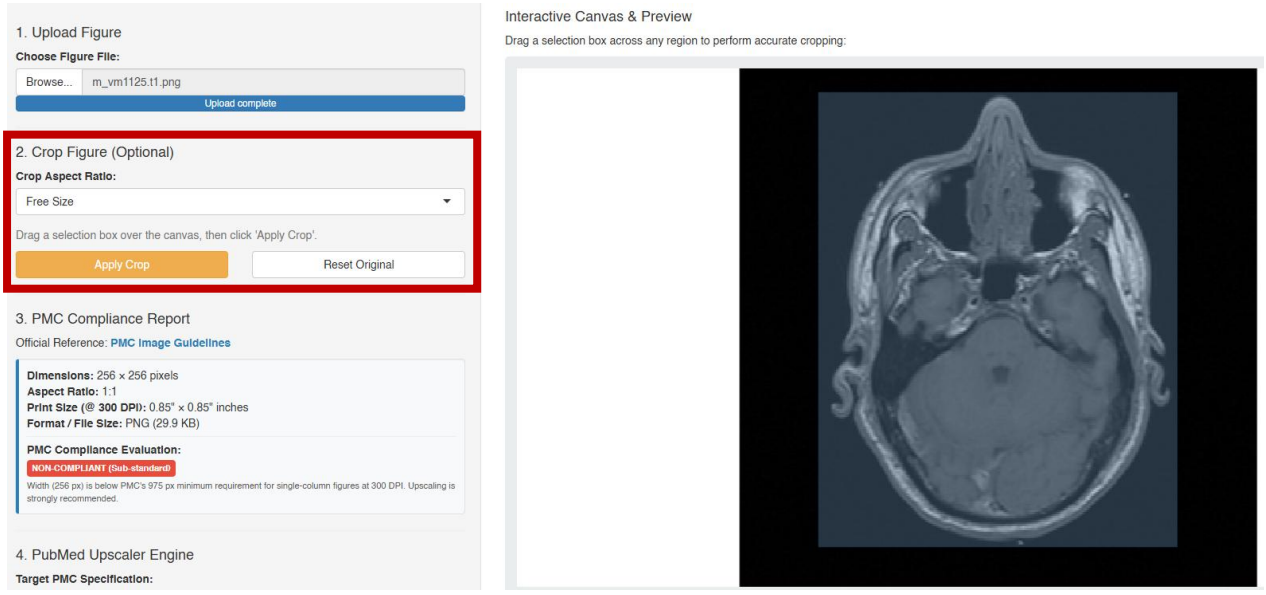
1. Upload Figure

Click Browse to upload your source file



2. Crop Figure

Select a region and click on “Apply Crop”. The aspect ratio can be changed.



License: CC-BY 4.0

3. PMC Compliance Report

The app reports pixel dimensions, aspect ratio, print size, file size, and compliance.

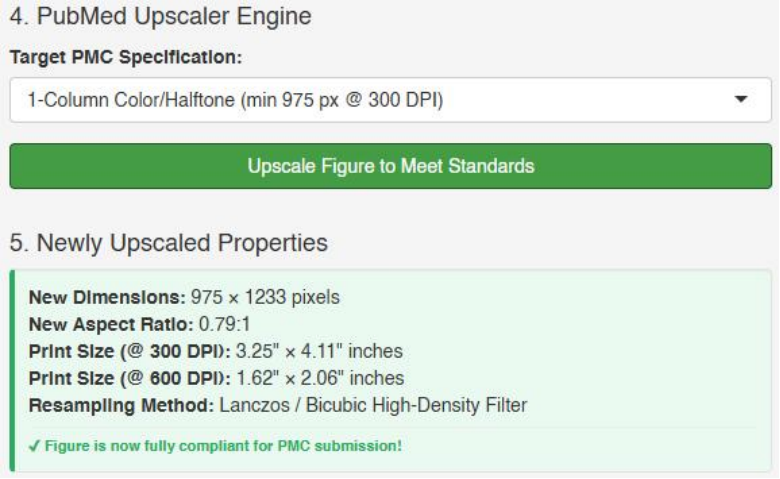


4. PubMed Upscaler Engine

If pixel density fails minimum PMC print thresholds, apply one-click pixel upscaling to achieve the required resolution.

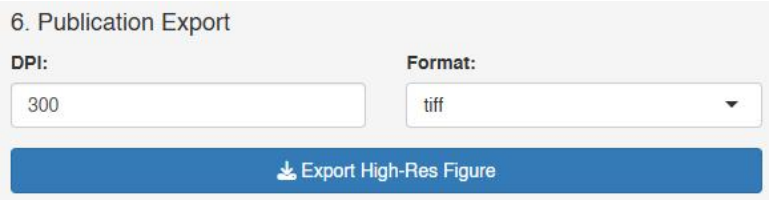
5. Newly Upscaled Properties

New image properties will be shown.



6. Publication Export

Select your preferred export format (TIFF is recommended for PMC deposition, or JPG/PNG/PDF) and click Export High-Res Figure.



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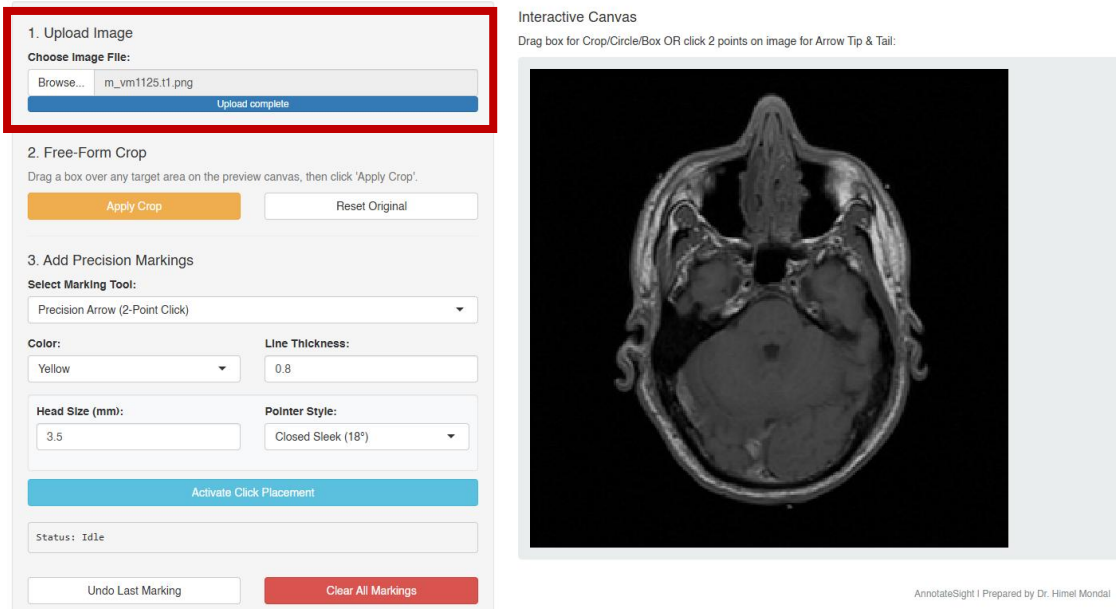
> Module 2: AnnotateSight

Audit raw figures against PMC guidelines, crop regions, and adjust pixel dimensions.

Functions:

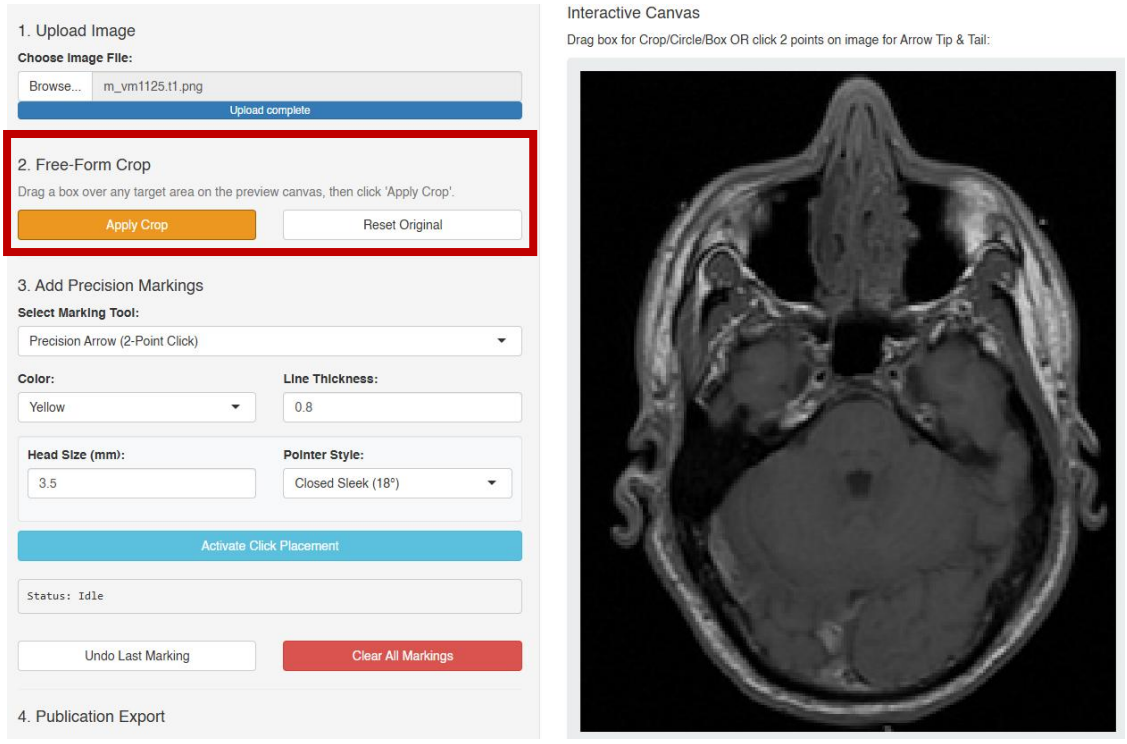
1. Upload Figure

Click Browse to upload your source file



2. Free-form Crop Figure

Select an area and “Apply Crop”. Only free-form cropping is available in this application.



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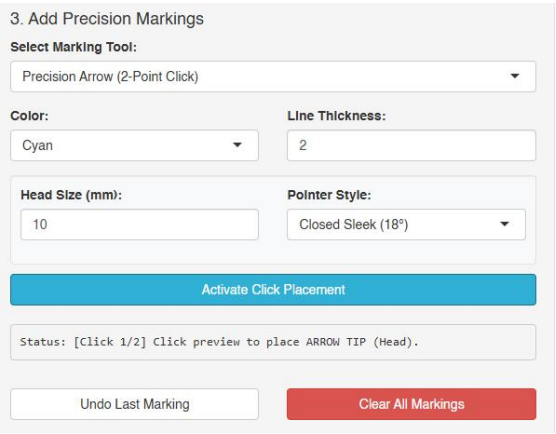
3. Add Precision Markings

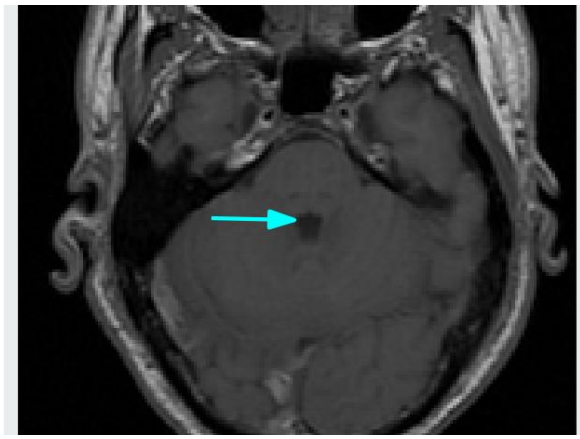
Choose stroke thickness and element color (pick high-contrast colors like yellow, white, or cyan).

Select Arrow, Rectangle, Circle, or Text.

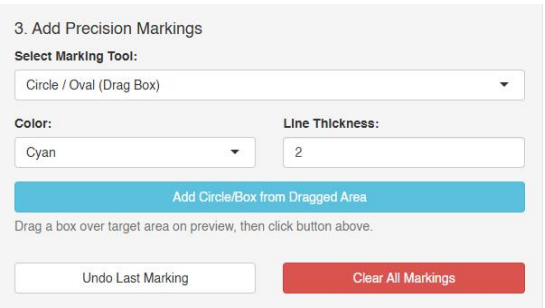


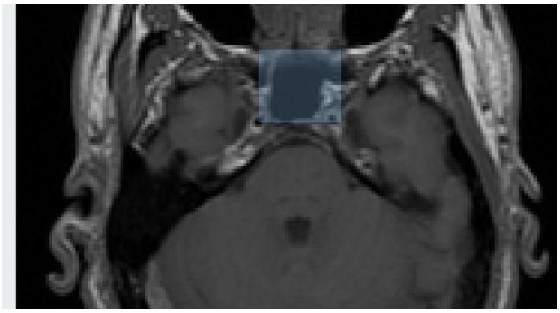
For arrow: Select “Precision Arrow”, click "Activate Click Placement," then click the start point (arrowhead) and endpoint on the figure.

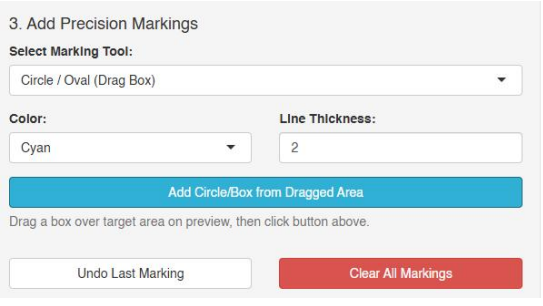




For circle and box: Select Circle or Rectangle, select the area, and then select “Add Circle/Box from Dragged Area”



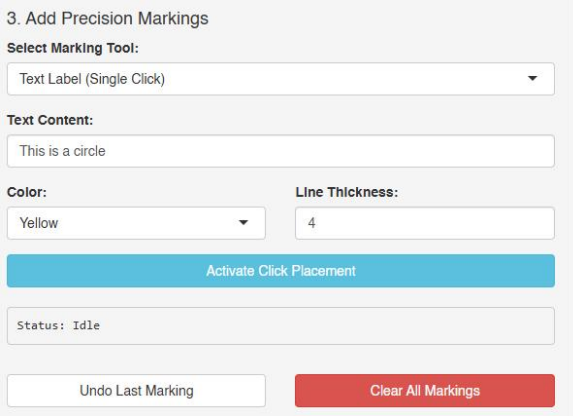


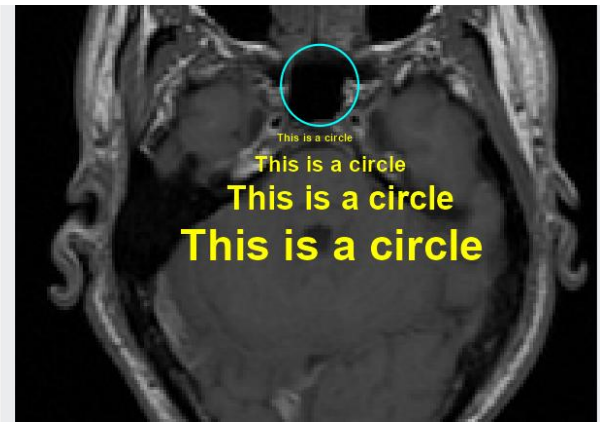




License: CC-BY 4.0

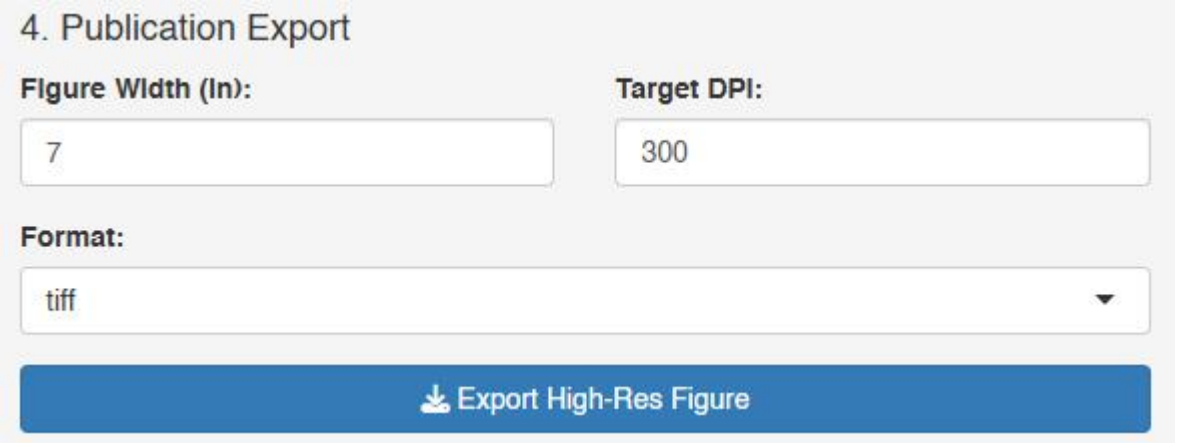
For text: Select “Text Label”, write text, select color and thickness, click “Activate Click Placement,” and click on the image; the text will be placed below and to the right of the click.





5. Publication Export

Select the preferred export format (TIFF is recommended for PMC deposition, or JPG/PNG/PDF) and click Export High-Res Figure.



License: CC-BY 4.0

> Module 3: PanelSight

Combine multiple discrete figures (e.g., pre/post scans, histological stains) into a standardized multi-panel grid with automated labelling.

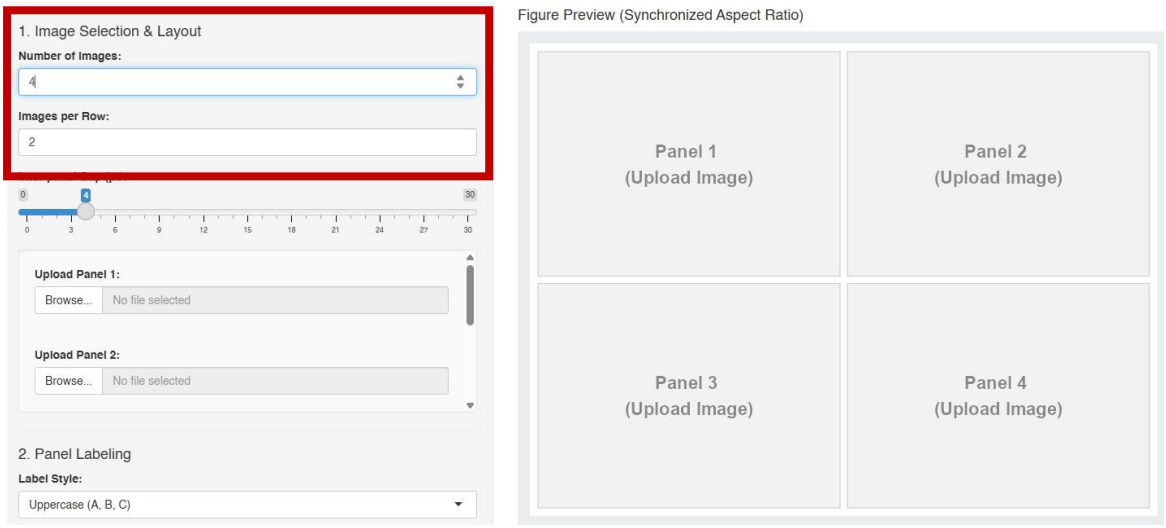
Functions:

1. Image Selection & Layout

Select the number of images (upload options will be updated according to the selected number) and images per row.

Upload multiple images simultaneously.

Grid dimensions automatically scale relative to the aspect ratio of the first uploaded file.



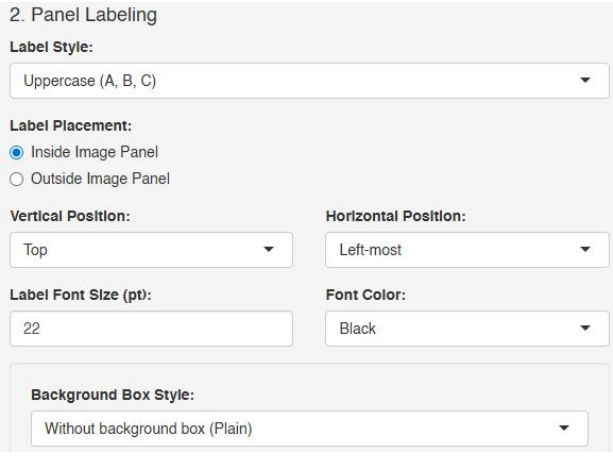
2. Panel Labelling

Case: Select Uppercase (A, B, C...) or Lowercase (a, b, c...).

Placement: Choose inside the image boundary or outside the frame.

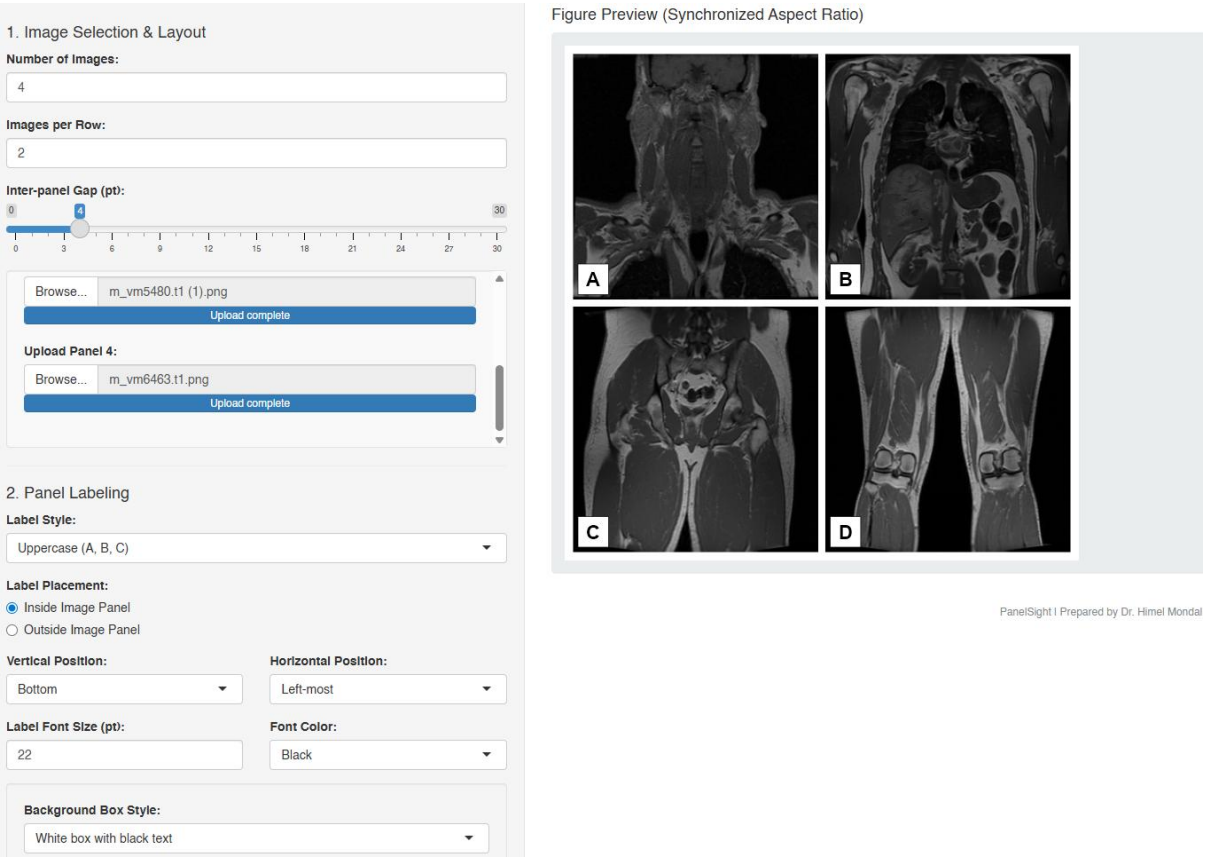
Position: Vertical (top or bottom), Horizontal (left, middle, right)

Box: Without box, white box (with black text), or black box (with white text)



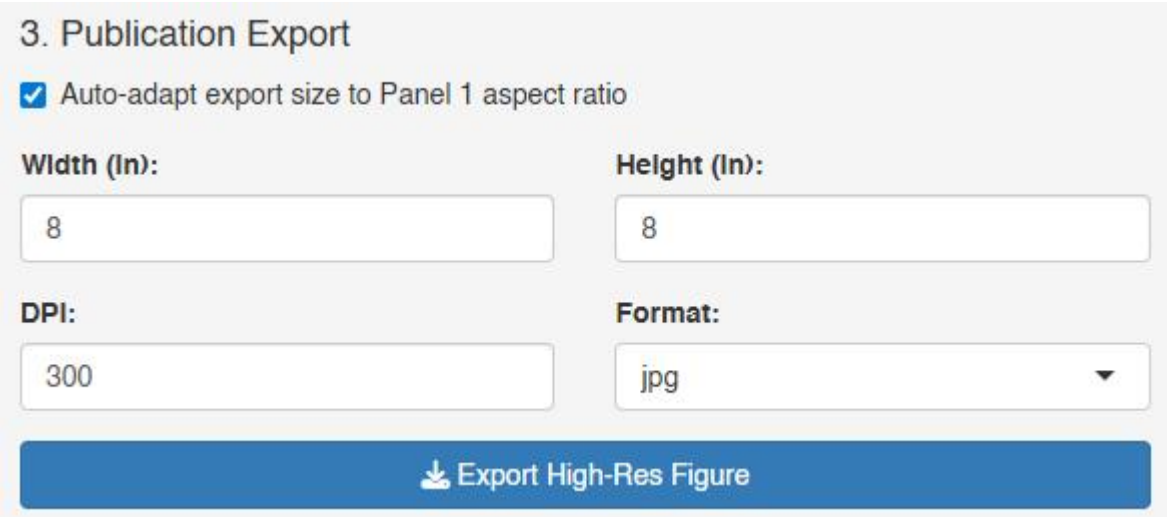
License: CC-BY 4.0

An example with an uppercase label, a white box, vertical bottom, and a horizontal left-most position with 22 font size



3. Publication Export

Select the preferred export format (TIFF is recommended for PMC deposition, or JPG/PNG/PDF) and click Export High-Res Figure.



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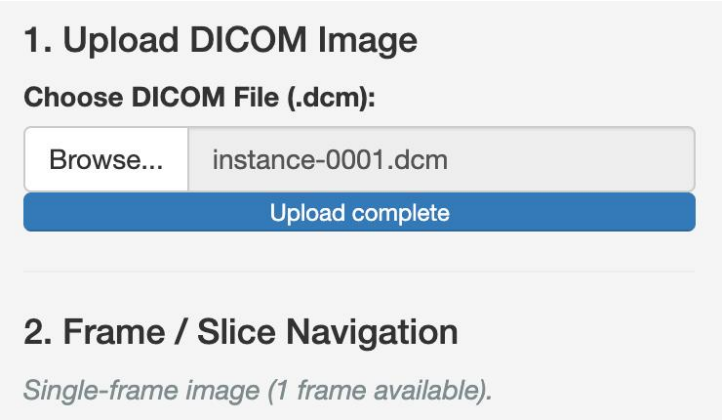
> Module 4: DICOMSight

Convert and process medical imaging files Digital Imaging and Communications in Medicine (.dcm) across CT, MRI, and USG modalities into publication-ready scientific figures.

Functions:

1. **Upload**

Upload .dcm file. It supports single-slice captures as well as multi-slice series up to 150 MB

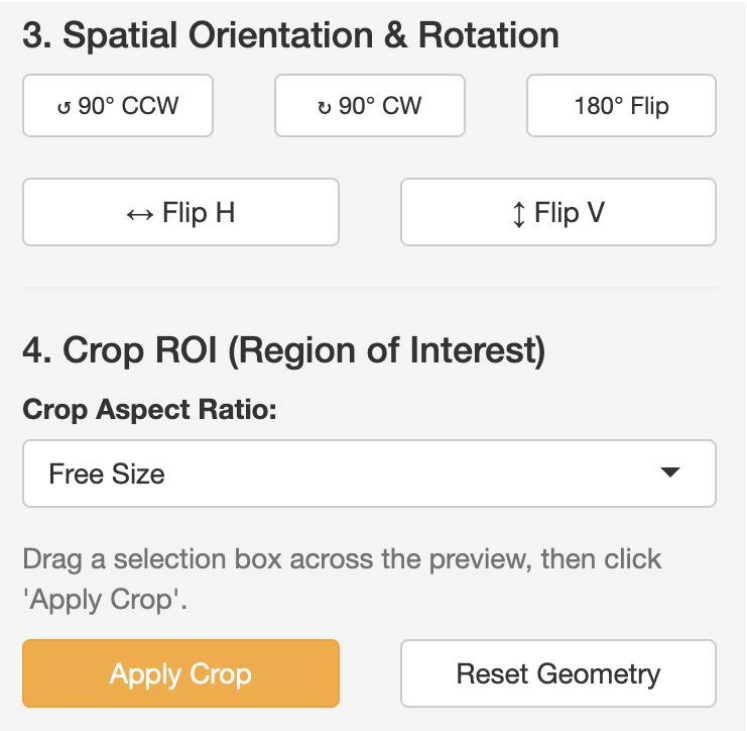


2. Frame / Slice Navigation

Dynamically generates an interactive slider when multi-frame or cine data is detected.

3. Spatial Orientation & Rotation

Rotate and flip scans



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4. Crop ROI

Drag a bounding box over the canvas to crop specific anatomical structures (region of interest).

Choose preset aspect ratios

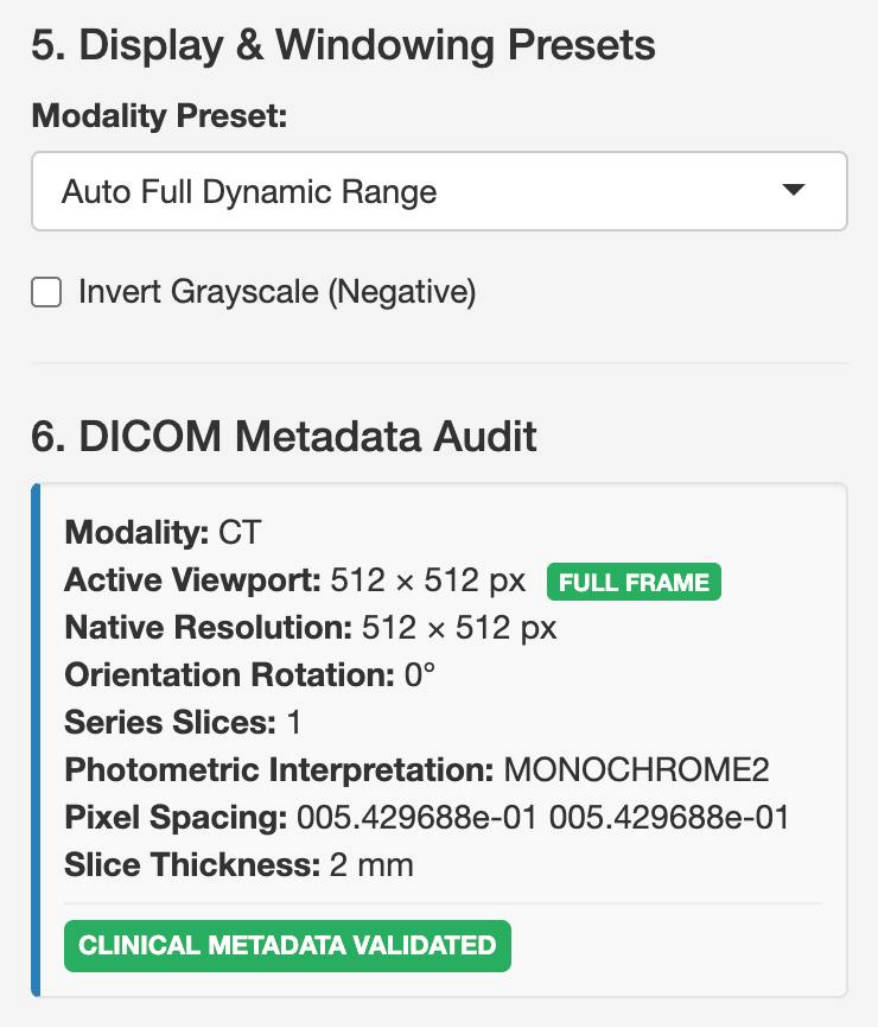
Apply crop

5. Display & Windowing Presets

It will be automatically selected according to uploaded image.

Can be changed according to necessity

Invert grayscale can be generated



6. DICOM Metadata Audit

Inspect key acquisition tags: modality, resolution, photometric interpretation, slice thickness, and others

License: CC-BY 4.0

7. Publication Export

Save in TIFF, PNG, JPEG, or PDF formats.

Customize physical print dimensions (width and height in inches) and export DPI for journal submission.



MRI images used in this user guide till page 10 are sourced from the Visible Human Project, US National Library of Medicine (NLM), https://www.nlm.nih.gov/research/visible/mri.html

Contact: Dr. Himel Mondal, AIIMS Deoghar, India (himel.physiology@aiimsdeoghar.edu.in)

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